

Capstone

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How can the **meaningful information limits** of generalized **AI** structures be inferred through the limits of **epistemology, constraint satisfaction, and information theory**?

1 Abstract

AI is an important part of the modern world, and as architectures are changing, and models are becoming increasingly larger, more general, and intelligent, it becomes important to have an intuitive model for the kinds of problems that AI can solve. This paper aims to provide such a framework for dismissing problems as impossible to solve with AI systems that is both workable and intuitive, incorporating a philosophic perspective, as well as a mathematical rigour.

2 Introduction

Artificial Intelligence, or AI, has grown significantly over the last decade with the rise of large language models and other foundation models [1]. These models are examples of neural networks, which learn from data rather than being programmed with exact task-specific behaviours [2]. Scaling studies of language models suggest that performance improves with additional parameters, data, and compute, while also making the cost of training and inference a central practical constraint [3, 4]. Although the exact internal mechanisms by which LLMs generalize across new tasks remain only partially understood [1], their shared structure of data, processing, and generation can still be used to frame arguments about the behaviour and limits of AI systems. This structural view of artificial intelligence motivates the use of constraint satisfiability problems, information theory, epistemology, and determinism when reasoning about the meaningful information limitations of generalized AI models.

Epistemology is the study of human knowledge [5]. It has also been used in recent work to analyse AI systems as black boxes and to reason about the limits of explanation, justification, and understanding in machine learning [6, 7]. Because data is a fundamental element of AI systems, epistemic questions about what is known, how it is learned, and what remains inaccessible apply directly to AI.

The different methods by which an AI system acquires and represents data also shape its structure, including the contrast between symbolic and connectionist approaches. (citation)

Constraint Satisfiability Problems, or CSPs, are a general framework for representing problems in terms of variables, domains, and constraints [8]. This abstraction makes CSPs useful for discussing classes of problems independently of the details of any particular solver [9]. CSPs have long been used in symbolic AI to represent relational structure in domain-specific problems while supporting general solving procedures [8, 9]. This style of representation supports mathematical arguments that may extend to AI systems when those systems can be modelled in CSP-like terms.

Information theory is a framework for quantifying information, communication, and uncertainty [10]. In machine learning, information-theoretic methods have been used to study how representations preserve, compress, and transform information during learning [11, 12]. Information theory therefore offers one way to analyse the internal behaviour of contemporary connectionist models [12]. Determinism is the view that events are fixed by prior conditions rather than arising independently. (citation) Although this remains debated across philosophy and physics, it provides a framework for asking whether future states of a system are in principle knowable from present information. (citation)

3 Background

3.1 Defining AI

Some cite AI as being any algorithm that aims to replicate human’s mental faculties, instead of their physical qualities. This definition, some say, (citation) is redundant since all algorithms are automating mental faculties of people. Other definitions aim to avoid redundancy by specifying how algorithms process data as prediction machines (citation). This provides a much better basis for understanding what AI is, some introductory courses to the field will describe AI as general function approximation algorithms (citation). This has been expanded in recent years as AI models have become successful in problems that were not previously understood as functions (citation), such as natural language processing. The ability to gain meaningful information moves beyond pure numeric function approximation has incurred the use of definitions of AGI, or artificial general intelligence, to differentiate simpler AI from more capable AI models. AGI can similarly be defined in many ways, but, as its name implies, is a generalization of AI, where all mental faculties of people are replicated, this was popularized as a concept by Alan Turing, as he describes the Turing test (citation), whereas a human interacts with an algorithm the human cannot differentiate if an output is the result of an algorithm or another human.

These conflicting definitions are now being shaped by current use of AI as the public are maintaining relationships with the field. This motivates the analysis

of these definitions for their applicability in supporting modern uses of AI. The aforementioned definition of AI as automation of mental faculties, in this context, gains meaning since it provides implied use cases for AI, as a tool to replace their mental faculties. The definition of AI as a prediction machine can be introduced to a member of the public to provide understanding of the mechanisms by which AI automates mental processes. The final definition of AI pertinent, here, is that of the general function approximator, where it again implies a certain use case, that it be used to regress or approximate functions, without the condition of knowledge of the structure of the function.

These standard definitions excel at providing intuitive understand for a lay person to understand what AI is, and how it can be used, but they fall short in the ability to capture the limitations of AI. This motivates the position here, that the definition for AI should be structural, instead of useful, and to provide an intuitive construction of this definition.

3.2 Epistemic Limitations

Knowledge has been defined as a justified true belief (JTB), although this has been debated by Gettier (citation), as it does not account for cases such as accidental knowledge. This process of knowledge acquisition has many perspectives. Two of these perspectives are that of the empiricists and rationalists. Philosophers such as Hume, argue that knowledge is gained through experience, giving us more evidence, thus increasing the likelihood or probability that what we believe is true, justifying it based on inference from our ability to sense reality (citation). Descartes, alternatively, in his meditations on first philosophy, that the only knowledge that we can have is that which is infallible, constructing his method of doubting to conclude that his ability to doubt is the only infallibility. These can be understood as different premises of the JTB theory, where Descartes argues that justification and truth can only come through rationality, Hume argues that justification and truth come from our experience of reality. These perspectives ask "what can one know?" in the context of their given reality. In AI problems, the reality must be provided,

As shown in thought experiments such as Newcomb's problem [13], future events are often treated as examples of the unknowable.

Knowledge, as defined here is not sufficient to describe the use case of AI models, since knowledge is specific to the agent that understands the knowledge. Let meaningful information, then, be defined as a valid collection or arrangement of signs that incur some meaning to a decoder. Where validity is defined as a constraint on the model, meaning is assumed to be captured in the signs of the domain¹, and signs are symbols that incur meaning beyond their literal shape (citation - semiotic theory).

¹In extensions of this theory, this assumption must be taken into account, that the problem might be solvable on some general AI model.

3.3 Defining Terms

Definition 3.1 (Constraint Satisfiability Problem). A constraint satisfaction problem (CSP) is a triple

$$S := \langle X, D, C \rangle,$$

where:

- $X := \{x_1, \dots, x_n\}$ is a set of variables x_i .
- $D := \{D_1, \dots, D_n\}$ assigns a domain D_i to each variable x_i .
- $C := \{c_1, \dots, c_m\}$ is a set of constraints c_i .
 Each constraint $c_j \in C$ is a pair $c_j = \langle S_j, R_j \rangle$, where $S_j \subseteq X$ is the scope and R_j is a set of valid tuples $R_j \subseteq \prod_{x_i \in S_j} D_i$.
- $S := \left\{ \mathbf{x} \in \prod_{\ell} D_{\ell} : \forall c, \mathbf{x}_{S_c} \in R_c \right\}, |S| > 0$ is the set of valid assignments $\mathbf{x}_k$.

Definition 3.2 (Admissible Pairs). Let D_{xy} denote the admissible pairs (x, y) compatible with S .

Definition 3.3 (Probability). Let $P_i(a)$ denote the probability $X_i = a$ or $p(X_i = a)$ and be defined as,

$$P_i(a) = \frac{|\{\mathbf{x} \in S : x_i = a\}|}{|S|},$$

assuming a uniform distribution.

Let $P_{ij}(a, b)$ denote $p(X_i = a, X_j = b)$, and be defined as,

$$P_{ij}(a, b) = \frac{|\{\mathbf{x} \in S : x_i = a, x_j = b\}|}{|S|}.$$

Definition 3.4 (Surprisal). The surprisal of an outcome a of random variable X is,

$$I_i(a) := -\log_2(P_i(a)),$$

where $p(x)$ denotes the probability of a .

Definition 3.5 (Entropy). The entropy of random variable X is,

$$H(X_i) := - \sum_{a \in D_i} P_i(a) \log_2(P_i(a)).$$

4 The model of AI

4.1 Constructing the model for AI

To justify the model of Input $\rightarrow$ Processing $\rightarrow$ Generation, the limitations established previously must be applied to each of these stages.

The empirical and rationalist perspectives motivate this paper's definition of AI.

To apply this to AI, we must construct the context or "reality" of our AI. This motivates the idea that we can suffice this context by constructing a domain of both our inputs and our generation.

1. Generation from information is impossible if there exists no relationship between the learned data and generated data.

Proposition 4.1. *Model claims*

1. *For all AI models, there exists an equivalent constraint satisfaction representation.*
2. *For all AI models, the relationship between generation and information is proportionately tractable to the similarity between the information and the generation.*
3. *For all AI models, there exists no equivalent process to generating the future to absolute detail at the current speed of time that is tractable.*

Proposition 4.2. *Processing requires that a constraint from x to y exists. For constraints between a tuple of variables T , a single equivalent constraint can be constructed of $T \cup U$ that is equivalent. Over constrained, if there exists no generation, for any of the input's domain the problem is invalid. Under constrained, if the entropy of the generation is not less than the entropy of the input, the problem is invalid*

4.2 The mathematical model for AI

Proposition 4.3. *Over constrained: $D_y(x) = \emptyset$. Under constrained: $|D_y(x)| = |D_y|$. Valid constraint effect: $0 < |D_y(x)| < |D_y|$.*

5 Conclusion

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